

6th Sem

2nd INTERNAL /

COMPUTER NETWORKS / PCC-CS602 / 6TH SEM-CSE / 02.05.2025 / FM: 15

1. What is framing? Discuss and differentiate between byte stuffing and bit stuffing? 1+4=5
2. What is the advantage of two dimensional parity over simple parity? Explain with suitable example. 5
3. "In Selective-Repeat ARQ, sender window size cannot be $>2^{m-1}$ ". Justify it with a suitable pictorial example. 5

Information Theory & coding Internal-II

$$C(x) = \overline{M(x)} G(x)$$

x^{n-k} $M(x) + P(x)$
R of $\frac{n-k}{G(x)}$

1.a) Write down the properties of cyclic code.

b) The generator polynomial of (7,4) cyclic code is $G(x) = x^3 + x + 1$. Find code vector for message 1010.
2+5

2. Write the algorithm of Hamming code with proper example. Write the expression of degree of BCH code and generator polynomial of BCH code. 4+4

JGEC/B.TECH/ CSE/HU-601/ 2025
Principle of Management

*Lack of
comm
Authority*

Full Marks: 15

Times: 45min

- 1) Discuss the various techniques of group decision making. Write common errors in decision making. Write advantage and disadvantage of bureaucratic management.

(5+5+5)

Delpni

Department of CSE Subject-PCC-CS 601 Internal -2 Full Marks 15 Date: 02/05/2025

A. Answer any two

- 1 i. Find out the three-address code for the expression $a = b * -c + d * -e$ 3
- ii. Write the quadruple for the above expression 3
2. Illustrate loop optimization with suitable example. 6
3. Explain in detail about peephole optimization. 6

B. Answer any two

- 1) What is basic block? What are the rules to find leader in basic blocks? 1+2
- 2). Write the differences between S-Attributed and L-Attributed SDD 3

freq. dist.
loop unrolling
loop tam

CSE/PEC-CS601A/2025-26
DATA WAREHOUSING & DATA MINING

Full Marks: 15

Times: 45 min

Answer all questions

- i) "Classification is a supervised learning but clustering is an unsupervised learning." Explain it. 2
- ii) What are the advantages and disadvantages of k-means algorithms? 3
- iii) Draw a decision tree from the training data set given in Table 1 using information gain. {*cheat* is the class label attribute}. 10

TID	Marital Status	Taxable Income	Refund	Cheat
1	Single	High	No	Yes
2	Married	High	No	No
3	Single	Low	No	No
4	Married	High	Yes	No
5	Divorced	Average	No	Yes
6	Married	Low	No	No
7	Divorced	High	Yes	No
8	Single	Average	No	Yes
9	Married	Low	No	No
10	Single	High	Yes	No

Table 1

1 x 2 3 4

3 2 x 2 3 4

Full Marks:15

[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/CSE/HU601/ 2025
2025

PRINCIPLE OF MANAGEMENT

Times:45MIN

1. What is corporate social responsibility? In what different ways can business organization discharge these responsibilities? Write function of management. 3+6+6

1st INTERNAL / COMPUTER NETWORKS / PCC-CS602 / 6TH SEM-CSE / 11.03.2025 / FM: 15

1. What are the different components of data communication? 3
2. Examine many forms of physical topologies, including their advantages and disadvantages. 6
3. Discuss about different addresses provided by the different layer of TCP/IP model with size and example. 6

PEC-CS602B Internal Question

Total marks:15

Time-45 Minutes

1. Define entropy. Consider a discrete memoryless source with a source alphabet $A = \{s_0, s_1, s_2\}$ with respective probs. $p_0 = 1/4$, $p_1 = 1/4$, $p_2 = 1/2$. Find the entropy of the source.

2+4

2. Using Shannon-Fano algorithm, find Entropy, efficiency for six messages occurring with probabilities, $1/24$, $1/12$, $1/24$, $1/6$, $1/3$, $1/3$.

4+3+2

Soft skills and interpersonal development- OEC CS601

INTERNAL EXAMINATION QUESTION PAPER

FULL MARKS: 15 ANSWER ALL QUESTIONS: $3 \times 5 = 15$

1. What are soft skills? How do you differentiate between hard skills and soft skills?
2. Explain the concept of a leader.
3. Why is it important to develop positive thinking and attitude?
4. What skills are essential for effective team work?
5. Define and explain the concept of time management in your own words.

JGEC/CSE/Class Test I/2025/EVEN
DATA WAREHOUSING & DATA MINING(PEC-CS601A)

Time: 45 minutes Full marks: 15

Answer question 1 and any one from the remaining.

10+5=15

1. Enumerate all frequent item-sets from the given database (Table-1) using Apriori Algorithm with minimum support count $S=3$. List all the candidate sets and large frequent item-sets for each iteration. Show any three strong association rules derived from 3-large frequent item set (L3).

conf: 60%

Trans. id	A	B	C	D	E	F	G	H	I	J	K
T100	x	x	x	✓	x	✓	✓	✓	✓	x	✓
T200	x	x	✓	✓	x	✓	✓	✓	✓	x	✓
T300	x	✓	x	✓	✓	✓	x	x	✓	x	x
T400	✓	x	x	✓	x	✓	✓	x	x	x	x
T500	x	✓	x	x	x	✓	✓	x	x	✓	✓

✓ means the item is purchased and x means the item is not purchased

Table - 1

2. Draw a lattice of cuboid for four dimensional (item, location, time, customer) data. Define ETL steps and discuss the importance of ETL steps in data mining.
3. Define dimension and fact table of a schema. Compare star scheme with snowflake schema. Draw a probable concept hierarchy for "location" dimension.

2+3=5

2+2+1=5

2-3: Huffman, Star
 -6 = Hamming
 : BCH
 ear

Compiler Design (PCC-CS601)

Date of Examination: 11.03.25 (Internal Exam-1) CSE Dept.

Full Marks: 15

Time: 45 Minutes

Attempt any three. (Each question carries 5 marks)

1. Compute FIRST and FOLLOW set for all the non-terminals of the following grammar.

$$S \rightarrow \underline{A}Ba \mid bCA$$

$$A \rightarrow cBCD \mid \epsilon$$

$$B \rightarrow CdA \mid ad$$

$$C \rightarrow eC \mid \epsilon$$

$$D \rightarrow bSf \mid a$$

2. Consider the following context free grammar

$$S \rightarrow SS + \mid SS * \mid a$$

d) Give a rightmost derivation for the string $aa+ a^*$.

e) Is the grammar ambiguous or unambiguous? Justify your answer.

f) Describe the language generated by this grammar.

3. Briefly describe first four phases of a compiler with examples.

4. Write an unambiguous grammar which generates valid arithmetic expressions containing only 3 operators, exponentiation (^), multiplication(*) and addition(+). The operators are right, left and left associative respectively and are mentioned in the order of higher to lower precedence level.